

Apadmi Data Engineering Interview Task

Introduction

Thank you for considering Apadmi as the next step in your career. Prior to the face to face interview we would like you to attempt the following task. The main aim of this exercise is to allow us to get more insight on how you approach Data Engineering problems and spark more technical discussions in the interview.

We expect this to take roughly 4 hours. Please prioritise documentation and your architectural reasoning over a perfectly finished dashboard showing the results.

Main Task

Stack Overflow Annual Developer Survey

Overview

The 2025 Developer Survey is the definitive report on the state of software development. In its fifteenth year, Stack Overflow received over 49,000+ responses from 177 countries across 62 questions focused on 314 different technologies, including new focus on AI agent tools, LLMs and community platforms.

You can find the data here: <https://survey.stackoverflow.co/>

Task

Apadmi would like to make sure the salary we are paying our developers remains competitive based on their skills and experience. Please transform this survey data into a dimensional model so that it is ready for analysts to use in reports. Additionally, please create a visualisation showing years of experience and salary across different programming languages. If there are any other interesting relationships in the data you would like to visualise feel free to present those as well.

Please be ready to present any visualisations you have made and talk us through how you have processed and stored the data. Additionally, please ensure your solution is reproducible (e.g., via Docker, a requirements.txt, or a clear setup script).

Additional Tasks

If you finish the main task and still have time you are willing to commit, we have a few additional tasks. Please do not attempt to actually complete the tasks, we would like you to give us a high level overview of how you would approach the task along with any “gotcha’s” to watch out for based on the type of data or the task.

NYC Taxi & Limousine Commission (TLC)

Overview

This dataset contains millions of trip records (pick-up/drop-off times, locations, fares, etc.) available in Parquet format.

You can find the data here:

<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

Task

Join the trip data with the 'Taxi Zone' lookup table and create a dashboard showing the highest-earning zones by hour of the day.

Assume the data is too large to fit into the memory of your laptop.

Github Archive

Overview

This dataset contains hourly archives of every public event on GitHub in a nested json file format

You can find the data here: <https://www.gharchive.org/>

Task

Ingest the last 24 hours of GitHub events. Identify the top 10 most 'active' repositories, defining 'active' based on a weighted score of commits and stars

OpenAQ (Air Quality Data)

Overview

This dataset consists of streams of PM2.5, NO2, and CO measurements from sensors worldwide.

You can find the data here: <https://openaq.org/>

Task

Build a pipeline that cleanses the sensor data, handles missing values, and provides a 'Daily Average Air Quality Index' per city